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Electronics and Communication Engineering

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EDUCATION

- **National Institute of Technology, Silchar** Assam, India
B.Tech in Electronics and Communication Engineering; CGPA: 8.96/10 Aug 2022 – Present

INTERNSHIP EXPERIENCE

- **Numaligarh Refinery Limited (NRL)** Onsite
Industrial Internship May 2025 – Jul 2025
 - **Industrial Automation:** Observed ABB Distributed Control Systems (DCS), PLCs, industrial instrumentation, and process automation workflows used in refinery operations.
 - **Process Monitoring:** Gained practical exposure to real-time monitoring, industrial communication networks, and safety instrumentation across refinery and utility plants.

PROJECTS

- **SmartRecon-GST**  PyTorch, LayoutLM, SBERT, Flask
AI Document Intelligence & GST Reconciliation Jun 2025
 - **Document AI:** Developed an AI-powered invoice processing pipeline using LayoutLM (DocVQA), Tesseract OCR, and Sentence-BERT embeddings for automated document extraction and semantic invoice reconciliation
 - **GST Validation:** Built an HSN-aware GST verification engine using official CBIC datasets, Question Answering models, and semantic search to automate compliance checking and reconciliation.
- **Stable Video Infinity (SVI)**  PyTorch, Diffusers, GGUF
Optimized Diffusion Model Inference Jan 2026
 - **Inference Optimization:** Optimized Wan 2.2 14B diffusion model using 4-bit GGUF quantization and aggressive GPU memory management, enabling inference on free-tier Google Colab (Tesla T4).
 - **Video Generation:** Implemented split inference, autoregressive video generation, and RIFE-based frame interpolation for long-form AI video synthesis with reduced VRAM requirements.
- **Video High-FPS Interpolator (RIFE)**  PyTorch, CUDA, Docker
AI Video Frame Interpolation Jan 2026
 - **Computer Vision:** Implemented a deep-learning-based video interpolation pipeline using RIFE and optical flow for recursive multi-frame generation supporting 2×, 4×, and 8× interpolation.
 - **Deployment:** Developed a Dockerized CLI with FFmpeg integration, automated model setup, and reproducible GPU-based execution.
- **Multi-Modal Image Fusion**  Stable Diffusion, CLIP, T5
Generative AI Image Synthesis 2025
 - **Image Generation:** Developed a multi-modal image generation pipeline using Stable Diffusion, CLIP Interrogator, and T5 summarization for prompt-guided image synthesis from multiple reference images.
 - **Deployment:** Built Flask and Gradio interfaces with GPU-accelerated inference, prompt optimization, and remote deployment using Ngrok and Google Colab.

TECHNICAL SKILLS

- **Languages:** Python, C++, C, SQL
- **Machine Learning:** PyTorch, Hugging Face Transformers, Sentence Transformers (SBERT), Diffusers
- **Computer Vision:** OpenCV, Tesseract OCR, LayoutLM, DocVQA, Optical Flow, RIFE
- **Data Processing:** MySQL, NumPy, Pandas, PDFPlumber, PDF2Image, Pillow
- **Generative AI:** RAG, Stable Diffusion, CLIP, Prompt Engineering, GGUF Quantization
- **Frameworks:** Flask, Gradio, Docker, Firebase
- **Tools:** Git, GitHub, Google Colab, Linux, VS Code

HONORS AND ACHIEVEMENTS

- **Neurathon 2025 Finalist:** Selected among the top teams in a national AI hackathon for developing Machine Learning solutions in document intelligence and medical diagnosis.
- **Competitive Programming:** CodeChef – 3*; solved 500+ algorithmic problems across *Codeforces*, *Leetcode*, and *HackerRank*, strengthening problem-solving and algorithm design skills.
- **Open Source AI Development:** Developed multiple end-to-end AI systems spanning Document AI, Computer Vision, Diffusion Models, OCR, and Video Generation with production-ready deployment pipelines.
- **AWS Certified:** Completed *Generative AI with Large Language Models* along with DeepLearning.AI specializations in Machine Learning and Generative AI.